



Expert Study on Metadata for Mapping Reuse

This study investigates how metadata can help experts decide whether an existing **mapping artefact** can be reused in a new project. The research forms part of a PhD in Computer Science at **Trinity College Dublin**.

In this study, a **mapping artefact** means a technical specification used to transform, align, or connect data. Examples include an **interlinking rule**, an **RML mapping**, and an **ontology alignment**.

Goal of the experiment: The goal is to evaluate whether different metadata models provide enough context for an expert to understand a mapping artefact, judge its trustworthiness, and decide whether it could be reused as a starting point in another setting.

About the two metadata models:

Representation A: FAIR-IMPACT metadata model

This model provides a concise description of the mapping artefact itself, such as its identifier, source and target resources, creator, tool, context, and evaluation information when available.

Representation B: MMV lifecycle-based metadata model

This model describes the mapping artefact across its lifecycle. In addition to basic artefact information, it may include details about the purpose of the mapping, assumptions, design decisions, technical requirements, risks, testing, maintenance, and stakeholders.

What you will do**1. Review three reuse scenarios.**

Each scenario presents one mapping artefact from a different type of task:

- **Scenario 1:** Interlinking rule
- **Scenario 2:** RML mapping
- **Scenario 3:** Ontology alignment

2. Compare two metadata representations.

In each scenario, the same mapping artefact is described using two metadata models:

- **Representation A:** FAIR-IMPACT metadata model
- **Representation B:** MMV lifecycle-based metadata model

3. Make a reuse decision.

Your task is not to execute the mapping or inspect every rule in detail. Instead, please imagine that you have discovered this artefact while searching for reusable resources, and decide whether the metadata gives you enough information to reuse it as a starting point.

4. Complete a short questionnaire after each scenario.

You will be asked whether you would consider the artefact reusable, which representation better supported your decision, which one was easier to understand, and what information was most useful or missing.

Important: You are not being asked to judge the full technical correctness of the artefact itself. You are being asked whether the metadata description gives you enough information to make an informed reuse decision.

Please note: Please base your responses only on the information shown in the study pages. External searching or additional resources are not required.

Estimated duration: approximately **15–20 minutes** in total.

For questions about the study, please contact salzahra@tcd.ie.

Proceed

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